

University of Information Technology & Sciences (UITS)
Faculty of Science and Engineering
Department of Computer Science and Engineering
Program: B.Sc. in CSE
Term Final Examination, Spring 2025
Course Title: Coordinate Geometry Linear Algebra and Vector Analysis
Course Code: MATH0541211

Marks: 50

Time: 3(three) hours

(Answer all questions)

- | Q.No. | Questions | Marks |
|-------|---|-------|
| 1. | <p>a) Solve the following system of linear equations by using Gauss Elimination Method: [05]</p> $\begin{aligned} 3x_1 - x_2 + x_3 &= -2 \\ x_1 + 5x_2 + 2x_3 &= 6 \\ 2x_1 + 3x_2 + x_3 &= 0 \end{aligned}$ <p>b) Define consistent and inconsistent. Solve the following linear equations with the help of determinant (Cramer's rule) [05]</p> $\begin{aligned} x - 2y + 3z &= 11 \\ 2x + y + 2z &= 10 \\ 3x + 2y + z &= 9 \end{aligned}$ | |
| 2. | <p>a) Define Hermitian and Orthogonal matrix. Prove that [04]</p> $A = \begin{pmatrix} 2 & 2-3i & 3+5i \\ 2+3i & 3 & i \\ 3-5i & -i & 5 \end{pmatrix}$ is Hermitian <p>b) If $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{pmatrix}$ are two matrices. Then find AB^{-1} [6]</p> | |
| 3. | <p>a) Define Plane. Find the equations to the planes, [10]</p> <p>i) Passing through the lines of intersection $2x - y = 0$ and $3x - y = 0$ and is perpendicular to the plane $4x + 5y - 3z + 1 = 0$.</p> <p>ii) Which is parallel to the plane $3x + 2y + z = 7$ and passing through the point $(1, 2, 2)$.</p> | |
| 4. | <p>a) Find constant a, b and c so that a irrotational vector $\mathbf{V}$ is given by [03]</p> $\mathbf{V} = (4x - 3y + az)\hat{i} + (bx + 3y + 5z)\hat{j} + (4x + cy + 3z)\hat{k}$ <p>b) Suppose $\mathbf{A} = 2yz\hat{i} - x^2y\hat{j} + xz^2\hat{k}$, $\mathbf{B} = x^2\hat{i} + yz\hat{j} + xy\hat{k}$ and $\phi = 2x^2yz^3$. Find $(\mathbf{A} \times \nabla)\phi$ and $(\mathbf{B} \cdot \nabla)\mathbf{A}$ at the point $P(1,1,1)$. [04]</p> <p>c) Find the projection of $\mathbf{A} = 8\hat{i} - 2\hat{j} + \hat{k}$ on the vector $\mathbf{B} = \hat{i} - \hat{j} + \hat{k}$. [03]</p> | |

5.37 a) Suppose $A = (2y + 3)\hat{i} + xy\hat{j} + (yz - x)\hat{k}$. Evaluate $\oint A \cdot d\mathbf{r}$ along the following [03]

paths C:

$$x = 2t^2, y = t, z = t^3 \text{ from } t = 0 \text{ to } t = 1.$$

5.31 b) Let $A = t\hat{i} - 3\hat{j} + 2t\hat{k}$, $B = \hat{i} - 2\hat{j} + 2\hat{k}$ and $C = 3\hat{i} + t\hat{j} - \hat{k}$ then evaluate [04]

$$\int_1^2 A \cdot (B \times C) dt$$

c) A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = -t - 5$ where t is [03]

the time. Find the components of its velocity and acceleration at time $t = 1$ in the

direction $\hat{i} - 2\hat{j} + 2\hat{k}$

direction cosine?